CartCheck

Demonstration and viva guide

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Use this guide to present the Excel workbook clearly. Start with the problem, show the actual inputs and formulas, make one small change, and explain the result.

Opening script

Good morning ma’am. My Data Analytics project is CartCheck. It analyses whether food delivery discounts still produce a saving after delivery, packaging, platform and other charges. I used 100 simulated orders and implemented the analysis in Excel with formulas, PivotTables, charts and data quality checks.

Five minute demonstration

Time	Open and show	What to explain
0 to 1 min	Dashboard	The final bill includes every recorded fee. Negative saving is retained.
1 to 2 min	Orders A8:O8	200 subtotal minus 60 discount plus 45 fees gives 185 final bill and 15 saving.
2 to 3 min	M8, N8, O8 formula bar	SUM adds fees. The other formulas calculate final bill and saving.
3 to 4 min	Fee Planner C10	Change 20 to 30. Final bill becomes 195 and saving 5. Restore 20.
4 to 5 min	Analysis then Data Quality	Show a native PivotTable, explain refresh and show zero issues.

Closing script

In the study, fees total INR 7,634 against discounts of INR 6,750. This produces negative net saving of INR 884. Forty orders still save money, two break even and 58 have negative saving. The dataset is simulated, so these are demonstration findings rather than claims about real platforms.

Show the inputs and formulas

Open CartCheck_Excel_Project.xlsx in Microsoft Excel. Orders A to K contain inputs, L to T contain bill calculations, and U and V contain review formulas. Click a calculated cell and read its formula bar. Formulas > Show Formulas displays worksheet formulas together.

Cell	Starting input
F8	Food subtotal 200
G8	Discount 60
H8	Delivery fee 20
I8	Packaging fee 10
J8	Platform fee 5
K8	Other charges 10

Four formulas to memorise

```
M8 =SUM(H8:K8)
N8 =F8-G8+M8
O8 =G8-M8
P8 =IFERROR(O8/F8,"n.a.")
```

Say: Extra charges add the four fees. Final bill subtracts the discount and adds the fees. Net saving subtracts fees from the discount. Net saving percentage divides the saving by the listed food subtotal.

Lookup and classification

```
L8 =VLOOKUP(C8,$W$8:$X$10,2,FALSE)
S8 =IF(O8>0,"Positive",IF(O8<0,"Negative","Break-even"))
```

FALSE means an exact code match. The dollar signs keep the mapping range fixed when the formula is copied down. The nested IF classifies positive, negative and zero results.

Safe live change

Use Fee Planner C10 for the demonstration. Its starting value is 20. Change it to 30, show G8 = 55, G9 = 195 and G10 = 5, then restore 20. This does not alter the study orders. The workbook formulas are the project code; you do not need a separate Python file.

Viva questions on the analysis

1 What problem does CartCheck solve

It explains how the full set of extra charges changes the saving from a food order discount.

2 What is one row in your dataset

One food delivery order with its date, platform, subtotal, discount and fee components.

3 Where did the data come from

It is a simulated teaching dataset of 100 orders with fictional platform names.

4 What is net saving

Discount minus extra charges, or equivalently food subtotal minus final bill.

5 What does negative net saving mean

The final bill is above the listed food subtotal because fees exceed the discount.

6 Why use average fees by platform

The platforms have 34, 33 and 33 orders. Averages compare fees per order rather than volumes.

7 Why are mean and median different

Several very high fee values raise the mean. The median is less sensitive to the upper tail.

8 What is the mode here

The most frequent delivery fee is INR 20, occurring in 15 orders. It is a delivery-fee mode, not an extra-charge mode.

9 How do you find high-fee orders

Calculate Q1 and Q3, find IQR = Q3 - Q1, and flag extra charges above Q3 + 1.5 times IQR.

10 Is a high fee necessarily incorrect

No. It is a review flag. The order remains in the totals and needs more billing detail for explanation.

Viva questions on Excel and scope

11 What is a PivotTable

An Excel summary that groups records and applies calculations such as count, sum or average.

12 Which three PivotTables did you create

Average extra charges by platform, sum of final bills by month, and count of orders by saving outcome.

13 Why must you refresh a PivotTable

It uses cached source data. Refresh reads the updated orders and rebuilds the summary.

14 What do SUMIF AVERAGEIF and COUNTIF do

They sum, average and count records that match a specified condition.

15 What is a named range

A readable name for a cell range. FoodSubtotal refers to Orders F8:F107 and is used in Analysis I26.

16 How did you handle invalid inputs

Dropdowns and numeric validation restrict entry. Row checks and Data Quality identify missing values, duplicate IDs, mapping issues and invalid bills.

17 Why is the overall rate a ratio of totals

Total net saving divided by total subtotal weights the result by order value. A simple average of order percentages answers a different question.

18 Which platform is cheapest

The dataset is synthetic and baskets differ, so it cannot establish a cheapest real platform. DineDrop only has the highest average net saving in this study.

19 Did you use NLP deep learning or forecasting

No. The current project is Excel descriptive analytics and a deterministic fee calculator.

20 What would you improve next

Use consented anonymised bills with matched baskets and consistent fee definitions, then maintain ranges for a larger dataset.

Submission checks and quick fixes

Before the presentation

- Download the workbook and open it in Microsoft Excel. Keep the report PDF ready.
- Check that the six sheets and six dashboard charts are visible.
- Practise clicking Orders M8, N8, O8, L8 and S8 to explain the formula bar.
- Restore Fee Planner C8:C13 to 200, 60, 20, 10, 5 and 10. Restore target C15 to 20.
- Select Data > Refresh All if you edited Orders.
- Confirm Data Quality C6:C14 contains nine zero values.
- Keep the dataset description honest: 100 simulated records and fictional platforms.

If a chart does not change

If you edited Orders, refresh all PivotTables. The first three charts depend on their result cells. If you edited Fee Planner, the main dashboard should stay unchanged because its dataset is independent.

If Excel shows a value instead of code

Click the cell and look at the formula bar. To display all worksheet formulas, choose Formulas > Show Formulas. Click it again to return to normal results.

If the mode coverage check is not zero

A newly entered delivery fee may be absent from Analysis H30:H40. Extend the frequency list, update its formulas and the mode lookup range, and confirm the counts cover all orders.

If asked about 60 marks

Show evidence of the problem definition, input design, Excel formulas, PivotTables, statistical analysis, charts, testing and interpretation. The department decides the marking rubric. Do not promise a score or claim SQL, Tableau or AI features that are absent from this workbook.

One sentence to remember

CartCheck measures the saving that remains after all recorded delivery-related fees are deducted from a discount.